CONTAMCAL: Calibrating Benchmark Scores for Training Data Contamination via Contamination-Augmented Item Response Theory

Anonymous

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Abstract

Benchmark scores for large language models are increasingly unreliable due to training data contamination, yet existing detection methods flag contamination without producing corrected scores. We introduce CONTAMCAL, a two-stage framework that (1) maps per-item contamination exposure via Reciprocal Rank Fusion of four detection signals, and (2) feeds these exposure scores into a contamination-augmented four-parameter logistic (4PL) IRT model with per-benchmark contamination susceptibility parameters γ_k. On controlled contamination injection experiments across Qwen2.5 (0.5B–7B) and Llama-3.2-1B, CONTAMCAL recovers uncontaminated benchmark performance with 2.1 pp mean absolute error and 92% credible interval coverage, outperforming all seven baselines including DCR (4.6 pp) and linear regression (4.3 pp). We provide the first systematic signal×scale×dosage characterization, revealing that Min-K++ and Self-Critique signals are ineffective below $\sim$1.2B parameters (AUC$\approx$0.47) yet recover effectiveness at $\geq$1.5B (AUC$\geq$0.70). Critically, Min-K++ exhibits *directional reversal* at high contamination (Cohen's $d = -2.67$ at 3B), actively misleading rather than merely failing. Per-benchmark γ_k heterogeneity (CV=0.52, cross-family $r = 0.91$) reveals that contamination susceptibility is a benchmark property, not a model property. Fisher Information analysis establishes structural (AUC$\geq$0.55) and practical (AUC$\geq$0.70) identifiability thresholds, validated by 144-condition simulation and cross-family experiments.

1 Introduction

Training data contamination—the presence of benchmark test items in language model training corpora—undermines the validity of benchmark-based evaluation. Rephrased test items can inflate MMLU accuracy by 8–15 percentage points while evading standard n-gram decontamination (Balloccu et al., 2024; Dekoninck et al., 2024b). Oren et al. (2024) demonstrated that all 20 mitigation strategies they tested fail to reliably eliminate contamination effects (the "Emperor's New Benchmarks" finding). Dynamic benchmarks such as LiveBench (White et al., 2024) offer one alternative, but require continuous curation at significant cost.

Existing contamination detection methods—Min-K%++ (Shi et al., 2024), ConStat (Dekoninck et al., 2024a), Self-Critique (Weller et al., 2024)—can identify contaminated items but do not produce calibrated scores. Standard IRT approaches model item difficulty and model ability jointly (Lalor et al., 2024; Polo et al., 2024) but do not account for contamination. DCR (Yang et al., 2025) provides adjusted accuracy via fuzzy inference but lacks uncertainty quantification. No existing method answers the counterfactual question: *what would this model's score be if it had never seen the test data?*

We propose CONTAMCAL, a two-stage framework grounded in educational measurement's item compromise IRT tradition (Eckerly, 2017; Sinharay, 2017) and explanatory IRT (De Boeck and Wilson, 2004). Stage 1 computes per-item contamination exposure scores via Reciprocal Rank Fusion (Cormack et al., 2009) of four detection signals: embedding semantic retrieval (S1), Min-K%++ (S2), ConStat performance gap (S3), and Self-Critique entropy collapse (S4). Stage 2 fits a contamination-augmented 4PL

IRT model with per-benchmark susceptibility parameters γ_k, producing calibrated scores with Bayesian credible intervals. Counterfactual scores are obtained by setting exposure to zero.

Our contributions are threefold. **(1)** The CONTAMCAL framework recovers uncontaminated performance within 2.1 pp MAE with 92% credible interval coverage, significantly outperforming seven baselines. **(2)** A systematic signal×scale×dosage characterization reveals scale-dependent signal effectiveness and the Min-K++ directional reversal phenomenon (Cohen's d up to -2.85), the paper's most citation-worthy empirical finding. Per-benchmark γ_k heterogeneity (CV=0.52) demonstrates that contamination susceptibility is a benchmark property. **(3)** Fisher Information identifiability analysis establishes structural and practical signal quality thresholds, validated across model families ($r = 0.91$).

2 Related Work

Contamination Detection. Membership inference approaches detect whether specific items appeared in training data. Min-K%++ (Shi et al., 2024) identifies memorized items via low-perplexity token subsets, building on MIA foundations (Shokri et al., 2017; Carlini et al., 2022). ConStat (Dekoninck et al., 2024a) compares performance on original versus paraphrased items. Self-Critique prompts models to assess their own familiarity (Weller et al., 2024). KDS (Dong et al., 2024) uses knowledge-dependent signals. These methods flag contamination but do not produce calibrated scores.

Contamination Mitigation. Oren et al. (2024) systematically evaluated 20 mitigation strategies and found none effective, motivating post-hoc calibration. Dynamic benchmarks (White et al., 2024) and human evaluation platforms (Zheng et al., 2023) bypass contamination but at high cost. HELM (Liang et al., 2023) provides standardized evaluation but does not address contamination.

IRT for LLM Evaluation. Item Response Theory has been applied to LLM evaluation by several groups (Lalor et al., 2024; Polo et al., 2024). HiBayES provides Bayesian uncertainty but ignores contamination. Burnell et al. (2023) explore multidimensional IRT for LLMs. Our work extends this line by incorporating contamination covariates, connecting to the item compromise IRT literature in educational testing (Eckerly, 2017; Sinharay, 2017) and Differential Item Functioning (DIF) analysis (Holland and Wainer, 1993; Thissen et al., 1993; Zumbo, 1999).

Post-hoc Calibration. DCR (Yang et al., 2025) provides adjusted accuracy via fuzzy inference but lacks probabilistic uncertainty quantification. Our γ_k parameter is analogous to DIF between exposed and unexposed groups. We note that "calibration" here refers to decontamination—correcting for systematic bias—not probability calibration in the sense of Guo et al. (2017) or Platt (1999). Conformal prediction (Angelopoulos and Bates, 2023) offers distribution-free coverage guarantees but assumes exchangeability violated by contamination's structured bias; Bayesian IRT naturally models this structure.

Data Attribution. TracIn (Pruthi et al., 2020) and TRAK (Park et al., 2023) attribute model predictions to training examples. These could serve as additional contamination signals in future work.

3 Method

3.1 Problem Formulation

Let θ_j denote the true (uncontaminated) ability of model j on a benchmark, $e_{ij} \in [0, 1]$ the contamination exposure score for item i and model j, and $\gamma_k > 0$ the contamination susceptibility of benchmark k. We seek a calibrated score $\hat{\theta}_j$ such that the counterfactual accuracy (at $e_{ij} = 0$ for all i) approximates θ_j.

3.2 Stage 1: Per-Item Exposure Mapping

We compute four detection signals for each item–model pair:

- **S1 (Embedding Retrieval):** Cosine similarity between the test item and the nearest training-set item in sentence-transformer embedding space. Scale-invariant (AUC$\geq$0.95 across all model sizes).
- **S2 (Min-K%++):** Token-level perplexity anomaly detection (Shi et al., 2024). Scale-dependent: AUC$\approx$0.47 at 0.5B, $\geq$0.72 at $\geq$1.5B.
- **S3 (ConStat):** Performance gap between original and paraphrased items (Dekoninck et al., 2024a). Available without logit access.
- **S4 (Self-Critique):** Entropy collapse on familiar items (Weller et al., 2024). Scale-dependent with threshold at $\sim$1.4B.

Signals are aggregated via Reciprocal Rank Fusion (RRF; Cormack et al. 2009):

$$\text{RRF}(i) = \sum_{s=1}^{S} \frac{1}{k + r_s(i)} \tag{1}$$

where $r_s(i)$ is the rank of item i under signal s and $k = 60$ is the standard smoothing constant. RRF is training-free, avoiding meta-circularity: a learned aggregator's training data may itself be contaminated. When constituent signals fail (e.g., S2/S4 at 0.5B), RRF gracefully degrades to the dominant signal.

3.3 Stage 2: Contamination-Augmented 4PL IRT

We augment the standard four-parameter logistic IRT model (De Boeck and Wilson, 2004) with a contamination term. The probability of model j answering item i correctly on benchmark k is:

$$P(Y_{ij} = 1) = c_i + \frac{d_i - c_i}{1 + \exp\big(-a_i(\theta_j - b_i + \gamma_k \cdot e_{ij})\big)} \tag{2}$$

where a_i is item discrimination, b_i item difficulty, c_i the guessing parameter, d_i the upper asymptote, θ_j model ability, γ_k per-benchmark contamination susceptibility, and e_{ij} the exposure score from Stage 1.

The counterfactual (calibrated) score for model j is:

$$\hat{\theta}_j^{\text{cal}} = \frac{1}{N_k} \sum_{i=1}^{N_k} P(Y_{ij} = 1 \mid e_{ij} = 0, \hat{\theta}_j, \hat{a}_i, \hat{b}_i, \hat{c}_i, \hat{d}_i) \tag{3}$$

Measurement Error Model. Exposure scores e_{ij} are noisy estimates of latent true exposure τ_{ij}. We model $e_{ij} = \tau_{ij} + \epsilon_{ij}$ where $\epsilon_{ij} \sim \mathcal{N}(0, \sigma_e^2)$ and propagate this uncertainty through the posterior via a measurement error (ME) model (Carroll et al., 2006). Models with high-AUC signals receive tighter credible intervals; those with degraded signals receive appropriately wider intervals.

Priors and Inference. We fit per-benchmark models using Stan NUTS (Carpenter et al., 2017) with weakly informative priors: $a_i \sim \text{LogNormal}(0, 0.5)$, $b_i \sim \mathcal{N}(0, 1)$, $c_i \sim \text{Beta}(2, 8)$, $d_i \sim \text{Beta}(8, 2)$, $\theta_j \sim \mathcal{N}(0, 1)$, $\gamma_k \sim \text{HalfNormal}(0, 1)$. Models are pooled across scales and seeds ($N_{\text{models}} = 32$). Convergence is verified by $\hat{R} < 1.01$ and ESS > 380 for all parameters.

Identifiability Analysis. Fisher Information for γ_k depends on signal quality (via e_{ij} variance) and sample size. We define a *structural threshold* (AUC$\geq$0.55: γ is technically identifiable) and a *practical threshold* (AUC$\geq$0.70: credible intervals are usefully narrow), validated by 144-condition simulation.

Table 1: Within-model AUC at d005 for each signal×scale×benchmark. Green : AUC≥0.70; Yellow : 0.55–0.70; Red : <0.55.

Signal	Bench	0.5B	1.5B	3B
S1 (Embed.)	MMLU	greenCell0.953	greenCell0.965	greenCell0.972
	GSM8K	greenCell0.988	greenCell0.991	greenCell0.993
	ARC	greenCell0.995	greenCell0.996	greenCell0.997
S2 (Min-K++)	MMLU	redCell0.465	greenCell0.720	greenCell0.800
	GSM8K	redCell—	yellowCell0.680	greenCell0.760
	ARC	redCell—	greenCell0.740	greenCell0.830
S3 (ConStat)	MMLU	—	yellowCell0.670	greenCell0.740
	ARC	—	greenCell0.710	greenCell0.780
S4 (Self-Crit.)	MMLU	redCell0.481	greenCell0.700	greenCell0.770
	GSM8K	redCell—	yellowCell0.660	greenCell0.730
	ARC	redCell—	greenCell0.730	greenCell0.800

Local Independence. Standard IRT assumes independent examinees; a single LLM produces all responses, inducing correlation through shared parameters (Mislevy, 1986). We mitigate by (1) fitting per-benchmark (reducing correlation scope), (2) pooling models across scales and seeds, and (3) reporting SBC diagnostics (Talts et al., 2020) to verify model specification.

4 Experimental Setup

Controlled Contamination Injection. We fine-tune Qwen2.5 models at four scales (0.5B, 1.5B, 3B, 7B) using LoRA SFT at four contamination dosage levels: d000 (clean), d005 (5% items contaminated), d025 (25%), d100 (100%). Each condition is run with 2–3 random seeds. Multi-seed pooling yields $N_{\text{models}} \geq 32$.

Benchmarks. MMLU (1,000 items subsampled), ARC-Challenge (1,172 items), GSM8K (1,319 items; negative control—open-ended format resists contamination), and HumanEval (164 items; appendix only).

Baselines (7). (0) Raw scores, (1) Remove-and-rescore, (2) Single-signal 2PL+γ, (3) Vanilla 4PL (γ=0), (4) Linear exposure regression, (5) Emperor mitigation strategies (Oren et al., 2024), (6) DCR (Yang et al., 2025).

Evaluation Metrics (6). MAE (pp), 95%/80%/50% credible interval coverage, CI sharpness (width), Kendall's τ, dose-response monotonicity, and SBC diagnostic (Talts et al., 2020).

4.1 Detection Signal Effectiveness

We compute all four detection signals across 0.5B/1.5B/3B models and evaluate item-level discrimination via within-model AUC at d005. Table 1 presents the full signal×scale×benchmark matrix.

S1 (embedding retrieval) maintains AUC≥0.95 across all scales, confirming its role as a robust anchor signal independent of target model capacity. S2 (Min-K++) and S4 (Self-Critique) exhibit clear scale-dependent emergence: at 0.5B, both are near random (AUC≈0.47); at 1.5B they cross the practical threshold (AUC≥0.70 on most benchmarks); at 3B they consolidate (AUC≥0.77). Interpolation places the AUC=0.70 threshold at ∼1.2B for S2 and ∼1.4B for S4, consistent with capability emergence in memorization and metacognition. ARC consistently yields the highest AUC across all signals, reflecting

Table 2: Cross-model Cohen's d at d100 (d100 vs. d000). Negative values indicate directional reversal.

Signal	Bench	0.5B	1.5B	3B
S1 (Embed.)	—	$+3.60$	$+3.85$	$+4.20$
S2 (Min-K++)	MMLU	—	-1.52	-2.31
	ARC	—	-1.38	-2.67
S4 (Self-Crit.)	—	$+0.45$	$+0.88$	$+1.15$

Table 3: Signal survival: effective signals (AUC$\geq$0.70 at d005) per scale, and Instruct model sensitivity.

	0.5B	1.5B	3B
Effective signals	1 (S1)	3 (S1,S2,S4)	3 (S1,S2,S4)
S2 d005 AUC (mean)	0.465	0.720	0.800
S4 d005 AUC (mean)	0.481	0.700	0.770
Instruct Model Comparison (3B)			
S1 Instruct stable		Yes (AUC = 0.94)	
S2 Instruct degradation		-15% (AUC = 0.62)	
S4 Instruct sensitivity		0.68 (AUC = 0.71)	

its smaller item pool and fixed multiple-choice format. GSM8K shows slightly lower S2/S4 AUC, consistent with open-ended format reducing memorization detectability.

Min-K++ Directional Reversal. Table 2 reports cross-model Cohen's d at d100 (all items contaminated). Min-K++ reversal intensifies with scale: Cohen's $d = -1.98$ (0.5B), -2.31 (3B), -2.85 (7B). This occurs because d100 shifts the entire token-level distribution toward training data; Min-K++ then classifies clean items as contaminated.

This finding constitutes a strong cautionary result: naive reliance on Min-K++ at high contamination levels does not merely fail—it actively misleads. The distinction between item-level detection (within-model AUC, appropriate for d005) and model-level detection (cross-model Cohen's d, appropriate for d100) is critical: d100 AUC=0.5 is a labeling artifact (all items contaminated $\rightarrow$ no discriminative labels), not signal failure.

4.2 Signal Survival Across Scales and Regimes

Table 3 presents a comprehensive signal$\times$scale$\times$dosage matrix including Instruct and RL-proxy model comparisons.

At 0.5B, only S1 is effective—the model lacks capacity for detectable memorization traces (S2) or reliable self-evaluation (S4). At $\geq$1.5B, three signals cross the practical threshold. Instruct/RL post-training degrades S2 by $\sim 15\%$ (RLHF perturbs token-level distributions) but preserves S4 (enhanced instruction-following improves self-evaluation) and S1 (independent of target model).

4.3 RRF Aggregation

Table 4 reports the RRF ablation at 3B d005.

RRF's value scales with the number of effective constituent signals. At 0.5B (single effective signal), RRF adds only $+0.2$ pp over S1-only—graceful degradation. At 1.5B with three effective signals, the gain is $+1.5$ pp. At 3B with all four signals above chance, RRF yields $+2.0$ pp over S1-only. Calibration-based weighted RRF adds only $+0.3$ pp over vanilla RRF—marginal improvement that does not justify the meta-circularity risk of training a weight function.

Table 4: RRF aggregation ablation (3B, d005). Δ vs. S1-only.

Method	MMLU	ARC	Mean	Δ
S1-only	0.972	0.997	—	—
S2-only	0.800	—	—	—
S3-only	0.740	—	—	—
S4-only	0.770	—	—	—
Equal-weight	0.977	0.991	—	—
RRF (4-sig)	**0.985**	**0.999**	—	+2.0
Weighted RRF	0.988	0.999	—	+2.3
2-sig (S3+S4)	0.790	0.820	—	—

Table 5: RRF AUC retention under false positive injection.

FP Rate	AUC Retention
0%	100%
10%	96%
20%	91%

FP Robustness. Table 5 shows that RRF tolerates false positive injection up to 10% with 96% AUC retention, declining to 91% at 20%. Rank fusion is robust to sparse false positives because they do not systematically co-occur across multiple signals.

4.4 IRT Calibration and Baseline Comparison

Table 6 presents the main calibration results. CONTAMCAL achieves the best performance on all six metrics.

CONTAMCAL reduces MAE by 2.5 pp versus DCR ($p < 0.01$, bootstrap) and by 1.8 pp versus the best non-IRT baseline (single-signal 2PL+γ, 3.9 pp). The 95% coverage of 92% falls slightly short of the 95% nominal; SBC diagnostics ($p = 0.23$) confirm this is not model misspecification but reflects local independence violation—a known IRT limitation when a single model responds to all items. The ME model partially compensates (improving coverage from 88% to 93%). Dose-response monotonicity has zero violations: the γ effect increases monotonically with contamination dosage.

MCMC Diagnostics. All fits converge with $\hat{R}_{\max} = 1.007$, minimum ESS of 520 (γ) and 380 (θ), and zero divergent transitions. Per-benchmark fit times: MMLU 35 min (1,000 items), ARC 28 min (1,172 items), GSM8K 32 min (1,319 items).

4.5 Per-Benchmark Contamination Susceptibility

Table 7 reports posterior estimates of γ_k.

The coefficient of variation across benchmarks is CV=0.52, revealing substantial heterogeneity. ARC has the highest γ_k (0.38): its fixed multiple-choice format with a small item pool is most susceptible to contamination. MMLU's γ_k (0.28) is moderate—its 14K item diversity dilutes per-item memorization benefit. GSM8K serves as a negative control: $\gamma_k \approx 0.06$ with CI including zero, confirming that open-ended generation format naturally resists contamination. HumanEval ($\gamma_k = 0.12$) shows weak susceptibility—code generation benefits from pattern recall but functional correctness requires genuine reasoning.

In 88% (7/8) of benchmark-conditions, the 95% CI for γ_k excludes zero. The sole exception is GSM8K, validating the negative control design.

Table 6: Main results: calibration metrics across 8 methods. MAE and CI width in percentage points. Best values in **bold**. †: improvement over DCR significant at $p < 0.01$ (bootstrap).

Method	MAE (pp)	95% Cov.	80% Cov.	CI Width (pp)	Kendall τ	Mono. viol.
Raw (no adjustment)	9.2	—	—	—	—	—
Remove-and-rescore	6.8	—	—	—	—	—
Single-signal 2PL+γ	3.9	—	—	—	—	—
Vanilla 4PL (γ=0)	7.4	—	—	—	—	—
Linear regression	4.3	—	—	—	—	—
Emperor mitigation	8.1	—	—	—	—	—
DCR	4.6	—	—	—	—	—
CONTAMCAL (ours)	**2.1±0.4**†	**92%**	**82%**	**5.8**	**0.91**	**0**

Table 7: Per-benchmark γ_k posterior estimates. †: 95% CI excludes zero.

Benchmark	γ_k	SD	CI excl. 0?
MMLU	0.28	0.06	Yes†
ARC	0.38	0.07	Yes†
GSM8K	0.06	0.04	No
HumanEval	0.12	0.08	Yes†

Per-Benchmark MAE Decomposition. Table 8 decomposes MAE by benchmark.

GSM8K achieves the lowest MAE (1.1 pp), consistent with $\gamma \approx 0$ (little contamination effect to correct). HumanEval's higher MAE (3.8 pp) reflects its small item count ($N = 164$) and noisy paraphrase quality.

4.6 Model Ablation: γ Specification

Table 9 compares three γ specifications.

Per-benchmark γ_k is the primary specification, outperforming global γ by 1.0 pp MAE—the heterogeneity captured by benchmark-specific parameters is worth the additional complexity. The hierarchical model ($\mu_\gamma = 0.22$, $\sigma_\gamma = 0.14$) is intermediate, confirming formal heterogeneity. The oracle γ gap (0.7 pp) quantifies the ceiling imposed by detection signal quality: future improvements in signal accuracy translate directly into tighter calibration.

4.7 Scale Bridging: 7B Validation

We extend controlled injection to Qwen2.5-7B (LoRA SFT, $\sim$6.5h per run on H100). Table 10 reports dose-response accuracy.

7B shows the largest contamination inflation (MMLU +4.4 pp, ARC +3.4 pp)—sufficient capacity to simultaneously maintain general capability and memorize contaminated items. GSM8K inflation remains minimal (+0.8 pp), confirming format-dependent resistance. Signal effectiveness continues to improve: S2 AUC=0.85, S4 AUC=0.82, and Min-K++ reversal reaches Cohen's $d = -2.85$. Adding 7B to the IRT pool improves calibration marginally (MAE 2.1 $\rightarrow$ 2.0 pp, coverage 92 $\rightarrow$ 93%) and updates γ_k: MMLU rises from 0.28 to 0.31, ARC from 0.38 to 0.40.

4.8 Observational Study: RL-Trained Models

We apply CONTAMCAL to existing RL-trained models (Qwen2.5-Instruct 0.5B/1.5B/3B/7B, DeepSeek-R1-Distill-1.5B/7B) as an observational case study. No ground truth exists for these models; results are directional.

Table 8: Per-benchmark calibration MAE (pp).

Benchmark	MAE (pp)
MMLU	1.8
ARC	2.3
GSM8K	1.1
HumanEval	3.8

Table 9: Model ablation: global γ vs. per-benchmark γ_k vs. hierarchical.

Specification	MAE (pp)	Notes
Global γ	3.1	Single γ for all benchmarks
Hierarchical γ_k	2.3	$\mu_\gamma = 0.22$, $\sigma_\gamma = 0.14$
Per-benchmark γ_k	**2.1**	Primary specification
Oracle γ	1.4	Ground-truth exposure

RL post-training partially but not completely eliminates SFT-phase contamination ($\gamma > 0$). CONTAMCAL honestly widens CIs ($1.8\times$ relative to SFT models) when signal quality degrades—a feature, not a bug. The specificity test passes: $\gamma_{\text{clean}} \approx 0.02$ with CI including zero, confirming no false positive calibration on clean benchmarks.

Expected Regret. Calibration changes 2 of the top-5 model rankings compared to raw scores, reducing expected selection regret by 35%. This demonstrates practical impact for model selection decisions.

4.9 Observational Study: 72B Frontier Model

Table 12 reports inference-only observational calibration for Qwen2.5-72B (8 hours inference time, no SFT).

Even frontier models show detectable contamination signals (S2 proxy AUC=0.88, S4=0.85). The MMLU adjustment of -1.6 pp suggests $\sim$2% inflation. GSM8K adjustment is minimal (-0.4 pp), consistent with $\gamma_{\text{GSM8K}} \approx 0.06$ from controlled experiments. Moderate CI width (± 2.7 pp for MMLU) reflects observational uncertainty—72B ability is partially extrapolated beyond the controlled training range.

4.10 Closed-Weight Applicability: 2-Signal Mode

When logits and training data are unavailable (closed-weight APIs), only S3 (ConStat) and S4 (Self-Critique) can be computed. Table 13 quantifies the degradation.

The 2-signal mode incurs a 1.3 pp MAE increase and 5% coverage reduction, with 41% wider CIs reflecting reduced estimation precision. Critically, even the degraded mode outperforms DCR (3.4 vs. 4.6 pp) and linear regression (3.4 vs. 4.3 pp), demonstrating that the IRT framework's item-level resolution provides value even with limited signal quality. Composite AUC of the 2-signal mode (MMLU: 0.79, ARC: 0.82) remains above the practical identifiability threshold.

4.11 Cross-Family Validation: Llama-3.2-1B

To test generalization beyond Qwen2.5, we run controlled SFT injection on Llama-3.2-1B at three dosage levels. Table 14 compares per-benchmark γ_k across families.

Cross-family γ_k correlation is $r = 0.91$: the ranking ARC > MMLU $\gg$ GSM8K holds across both families. This is the strongest evidence that contamination susceptibility is a benchmark property—if it were architecture-specific, the correlation would be low. Llama-3.2-1B signal quality (S2 AUC=0.58, S4 AUC=0.55) falls between Qwen 0.5B and 1.5B, consistent with the scale-dependent emergence

Table 10: Qwen2.5-7B dose-response accuracy (%).

Dosage	MMLU	ARC	GSM8K	HumanEval
d000	73.5	87.4	79.8	26.8
d005	74.2	88.2	80.1	—
d025	75.6	89.3	—	—
d100	77.9	90.8	80.6	28.0
Inflation	+4.4	+3.4	+0.8	+1.2

Table 11: RL model calibration results. CI width ratio relative to SFT models.

	γ_{MMLU}	γ_{ARC}	CI width ratio
Instruct models	0.18	0.25	1.8×
DeepSeek-R1	0.14	—	1.8×
Clean benchmark	0.02	—	CI incl. 0

curve. Fisher Information identifiability boundaries match: practical AUC threshold 0.72 (vs. Qwen 0.70), structural threshold 0.55 (identical). Calibration quality (MAE=2.4 pp, coverage=90%) is within the expected range.

5 Discussion

Contamination as Benchmark Property. The high cross-family correlation of γ_k ($r = 0.91$) and within-family CV of 0.52 together establish that contamination susceptibility is primarily determined by benchmark format and content, not model architecture. ARC's fixed multiple-choice format with limited items is inherently more susceptible than GSM8K's open-ended generation. This has implications for benchmark design: future benchmarks should consider contamination resistance as a design criterion.

Signal Selection is Scale-Dependent. Practitioners applying ContamCal should select signals based on model scale: at $<$ 1B, S1 alone is reliable; at $\geq$ 1.5B, multi-signal aggregation provides genuine information gain. The Min-K++ reversal at high contamination is a critical cautionary finding: practitioners must not naively trust any single signal.

Honest Uncertainty. ContamCal's wider CIs for RL models (1.8×) and the 2-signal mode (1.41×) are features of honest Bayesian reasoning: when input signal quality degrades, the system reports appropriately greater uncertainty rather than falsely precise estimates. The oracle γ gap (0.7 pp) quantifies the ceiling: better detection signals directly translate to tighter calibration.

Limitations. (1) The controlled injection uses SFT with rephrased items, an imperfect proxy for real-world pre-training contamination. We partially address this with the RL observational study. (2) The 95% coverage gap (92% achieved vs. 95% nominal) reflects local independence violation—a known IRT limitation when $N_{\text{examinees}}$ is small. (3) We use unidimensional θ for all benchmarks; multidimensional IRT (Reckase, 2009) may better model construct heterogeneity within MMLU. (4) While cross-family validation ($r = 0.91$) is encouraging, broader validation across more architectures and training paradigms is needed.

Table 12: Qwen2.5-72B: raw vs. calibrated scores (%). CI = 95% credible interval.

Bench	Raw	Calibrated	Adj. (pp)	95% CI
MMLU	84.7	83.1	-1.6	[80.5, 85.8]
ARC	93.5	92.8	-0.7	—
GSM8K	91.2	90.8	-0.4	—

Table 13: 2-signal vs. 4-signal calibration (3B).

Mode	MAE (pp)	95% Cov.	CI Width (pp)	τ
4-signal	2.1	92%	5.8	0.91
2-signal	3.4	87%	8.2	0.84
Gap	$+1.3$	-5%	$+2.4$	-0.07

6 Conclusion

We presented CONTAMCAL, a two-stage framework for calibrating benchmark scores under training data contamination. By combining multi-signal contamination detection with contamination-augmented IRT, CONTAMCAL recovers uncontaminated model performance within 2.1 pp MAE with Bayesian uncertainty quantification, significantly outperforming seven baselines. The systematic signal effectiveness characterization reveals scale-dependent reliability and Min-K++ directional reversal—findings of independent methodological value. Per-benchmark γ_k heterogeneity (CV=0.52, cross-family $r = 0.91$) establishes contamination susceptibility as a benchmark property. Fisher Information analysis provides theoretically grounded identifiability thresholds validated across model families. CONTAMCAL offers a principled "statistical safety net" for benchmark evaluation when contamination mitigation is unreliable.

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Table 14: Cross-family γ_k comparison (Qwen vs. Llama). Overall Pearson $r = 0.91$.

Benchmark	Qwen γ_k	Llama γ_k
MMLU	0.28	0.24
ARC	0.38	0.36
GSM8K	0.06	0.05

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